California Polytechnic State University Pomona
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Intro to Microelectronics Laboratory

ECE 2200L
Report #2

LAB 2 – I-V Characteristics of the PN Junction Diode

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Presented to
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Introduction:

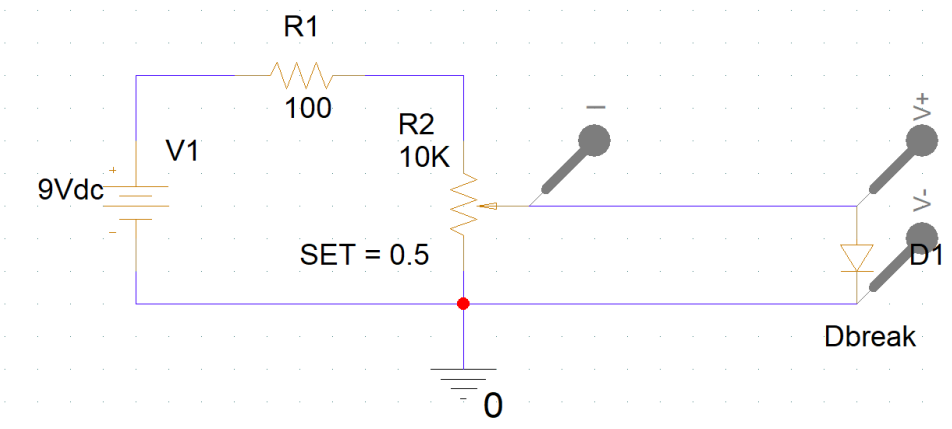
Our objective during this lab is to analyze the current-voltage relationship of two semiconductor PN diodes, one will be a Silicon diode and the other will be a Germanium diode. We also want to determine the reverse saturation current and diode ideality factor from the I-V characteristic plot from our gathered data. We will begin by observing the forward-biased relationship between the voltage over the diode for both diodes vs. various current values. We will get these measurements directly. Then we will observe the reverse-biased relationship between the voltage and the current over the diode. We will calculate current by getting the voltage across a 100k resistor, and then dividing the voltage by that resistance value to get current. Lastly, we will plot all of our graphs, and compare our data.

Parts List:

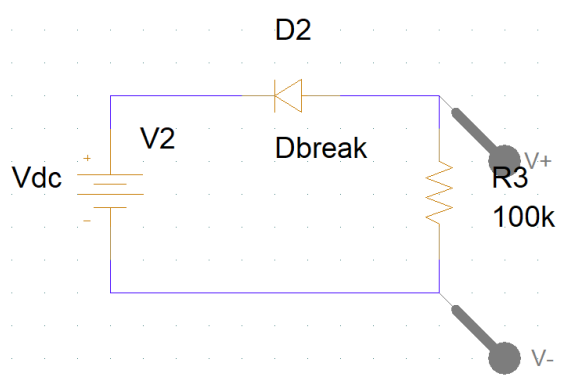
1. Resistors: 100 [Ω], 100 [kΩ]
2. Potentiometer: 10 [kΩ]
3. 1N4001 Silicon PN Diode
4. 1N270 Germanium Diode

Circuit Schematics:

Part 1:



Part 2:



Experimental Data:
Part 1:

I_D [mA]	0	0.2	0.5	1	2	4	8	16	20
V_D (Si) [mV]	0	538	577	610	639	670	706	742	752
V_D (Ge) [mV]	0	187	229	269	305	345	391	440	455

TABLE 1. Displays data collected during Part 1 of the lab.

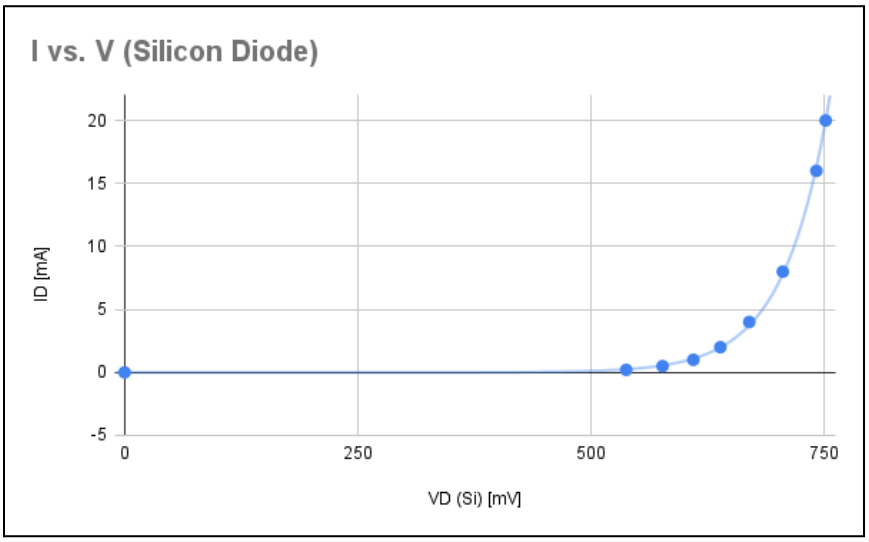


Fig. 1. Graphical depiction of data from Table 1.

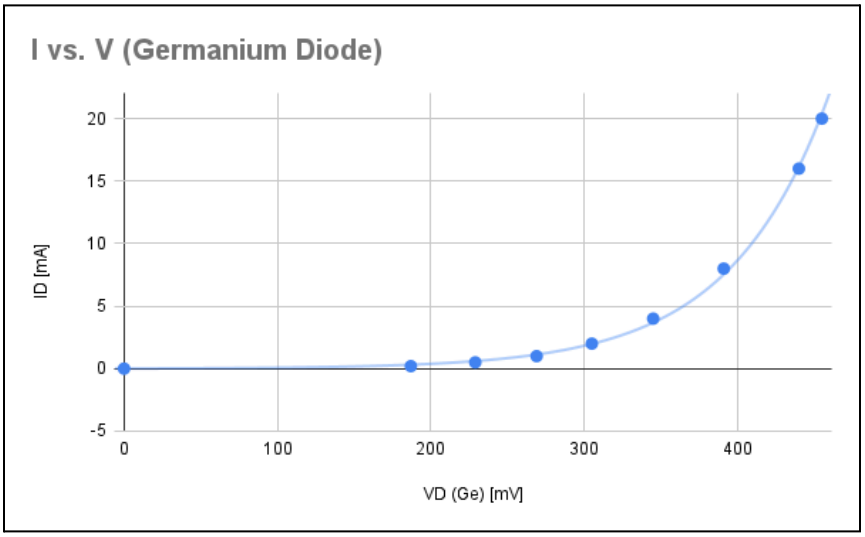


Fig. 2. Graphical depiction of data from Table 1.

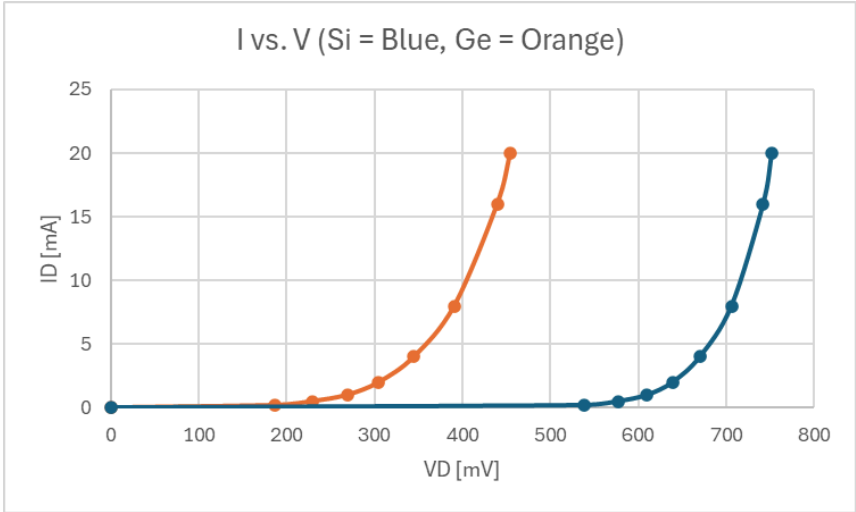


Fig. 3. Graphical depiction of data from Fig. 1 and Fig. 2.

Part 2:

V_D [V]	0	-2	-4	-6	-8	-10
V (Si) [mV]	0	0.18	0.22	0.26	0.29	0.32
I_D (Si) [nA]	0	1.8	2.2	2.6	2.9	3.2
V (Ge) [mV]	0	86.303	97.342	113.142	136.15	159.90
I_D (Ge) [μA]	0	0.863	0.973	1.131	1.362	1.599

TABLE 2. Displays data collected during Part 2 of the lab.

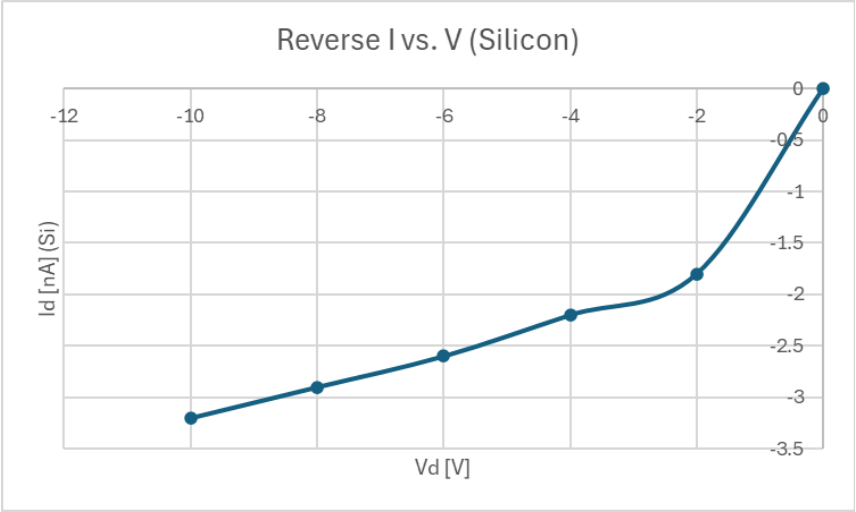


Fig. 4. Graphical depiction of data from Table 2.

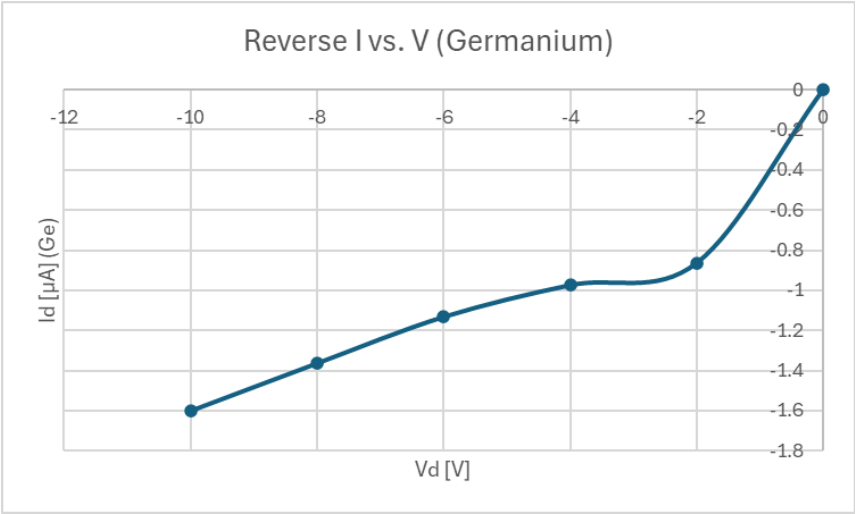
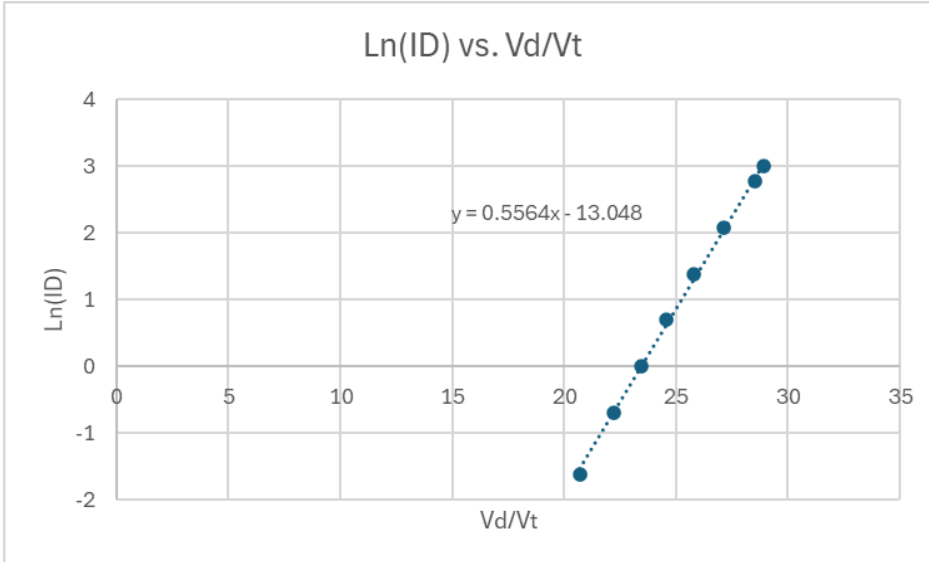


Fig. 5. Graphical depiction of data from Table 2.

Post-Lab:

Vd/Vt	20.69	22.19	23.46	24.58	25.77	27.15	28.54	28.92
ln(Id)	-1.61	-0.693	0	0.693	1.386	2.079	2.772	2.995



From the graph equation we can conclude that the y intercept is -13.048 and the slope is 0.5564. To find the ideality factor, n, we can use n = 1/slope. From this we get n = 1.8. To find Is, we can assume that -13.04 is equal to ln(Is). We can find Is by using $Is = e^{-13.048}$. From this we get Is = 2.15µA. This value is 10^3 times greater than the values we measured during the lab.

Conclusion:

After conducting our lab we were able to analyze the forward and reverse biased portions of the current-voltage relationship of two diodes. One of them is made of Silicon and the other is made of Germanium. We were also able to determine our ideality factor and reverse saturation current by plotting the natural log of ID vs Vd/Vt. When observing our Silicon and Germanium diodes in the forward biased portion, it was clear they were behaving as expected, but when they were compared to each other it was clear the Germanium diode was much more efficient since it has a smaller voltage drop. When observing our Silicon and Germanium diodes in the reverse biased portion, they also behaved as expected. When compared to each other, the Germanium diode had currents in the low micro region while the Silicon had currents in the low nano region. This means that the Silicon diode is more efficient in this case since it has less leakage. In conclusion, this lab was very informative and allowed us to confirm the voltage drop model of two different types of diodes.